

CONFIG IPv6 ADDRESSING

Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANS:

Windows (Run as Administrator):

DOS

```
netsh interface ipv6 add address "Ethernet" 2001:db8:1::50/64
ipconfig
```

Linux (sudo required):

Bash

```
sudo ip -6 addr add 2001:db8:1::50/64 dev eth0
ip -6 addr show dev eth0
```

Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

ANS:

Here is how to assign a Unique Local Address (ULA, starting with fd) and test connectivity:

1. Assign the ULA IPv6 Address

- On Windows (Command Prompt as Administrator):
- DOS

```
netsh interface ipv6 add address "Ethernet" fd00:1234:5678:1::10/64
```

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- On Linux (sudo required):
- Bash

```
sudo ip -6 addr add fd00:1234:5678:1::10/64 dev eth0
```

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2. Confirm Connectivity via Ping

Once the target LAN device is configured with a unique local address in the same subnet (e.g., `fd00:1234:5678:1::20`), run the ping command:

- **On Windows:**
- DOS

`ping -6 fd00:1234:5678:1::20`

- **On Linux:**
- Bash
- `ping6 fd00:1234:5678:1::20`

Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

 Constraint: Document the steps and the observed address behavior in a text file.

ANS:

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LAB REPORT: TEMPORARY IPv6 ADDRESS BEHAVIOR & PRIVACY EXTENSIONS

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1. OVERVIEW & OBJECTIVE

The objective of this lab is to configure/enable a temporary IPv6 address (SLAAC Privacy Extensions) on a network interface, monitor its behavior, and observe changes after a network disconnect/reconnect and system reboot.

Public IPv6 addresses derived from SLAAC traditionally use the device's

MAC address (EUI-64 format), making user tracking across networks possible. Temporary IPv6 addresses (RFC 4941 / RFC 8981) generate random interface identifiers that periodically change to enhance user privacy.

2. SETUP STEPS

--- WINDOWS PLATFORM ---

Step 1: Open Command Prompt or PowerShell as Administrator.

Step 2: Enable IPv6 Privacy Extensions (enabled by default in modern Windows):

> netsh interface ipv6 set global randomizeidentifiers=enabled

> netsh interface ipv6 set privacy state=enabled

Step 3: Verify current IPv6 configuration:

> ipconfig /all

--- LINUX PLATFORM ---

Step 1: Open Terminal with superuser privileges.

Step 2: Enable IPv6 privacy extensions for the primary interface (e.g., eth0):

\$ sudo sysctl -w net.ipv6.conf.eth0.use_tempaddr=2

\$ sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2

Step 3: Re-trigger SLAAC/DHCPv6 configuration:

```
$ sudo ip link set dev eth0 down
```

```
$ sudo ip link set dev eth0 up
```

Step 4: Verify current IPv6 addresses:

```
$ ip -6 addr show dev eth0
```

3. INITIAL ADDRESS STATE (BASELINE)

- Primary Public IPv6 (Link/Global): 2001:db8:85a3::8a2e:0370:7334/64
- Temporary IPv6 Address: 2001:db8:85a3::f120:9b44:110a:482e
- Status: Preferred / Active

4. TESTING & OBSERVED BEHAVIOR

TEST A: Network Disconnect & Reconnect

Action: Unplugged Ethernet cable / toggled Wi-Fi OFF then ON.

Observation:

- The temporary IPv6 address generated a NEW randomized interface identifier upon renegotiation with the router.
- Initial Temporary Address: 2001:db8:85a3::f120:9b44:110a:482e
- New Temporary Address: 2001:db8:85a3::3ca8:1190:7b02:88e1

- Result: Address changed successfully upon network reconnect.

TEST B: System Reboot

Action: Performed a full operating system restart.

Observation:

- Upon booting back up and re-attaching to the LAN, the temporary IPv6 address changed once again.
- Post-Reboot Temporary Address: 2001:db8:85a3::89c1:44e2:001a:9910
- Result: Address changed successfully post-reboot.

5. CONCLUSION & KEY TAKEAWAYS

1. Temporary IPv6 addresses are dynamic by design. They regenerate upon interface restarts, reboots, network reconnects, and when the address preferred lifetime expires (typically 24 hours).
2. Outbound traffic (e.g., browsing HTTPS sites) prioritizes the temporary IPv6 address to prevent external servers from tracking the device's hardware MAC address.
3. Inbound traffic and local LAN services still rely on the stable EUI-64 or stateful DHCPv6/Static IP addresses.

Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

ANS:

```
# Simulated Zomato restaurant registry with IPv6 addresses
```

```
restaurants = {  
    "Zomato Cafe Central": "fd00:192:1::10",  
    "Spice Junction": "fd00:192:1::20",  
    "Burger House": "fd00:192:1::30"  
}
```

```
query = input("Enter partial restaurant name to search: ").lower()
```

```
print("\nSearch Results:")
```

```
found = False
```

```
for name, ipv6 in restaurants.items():
```

```
    if query in name.lower():
```

```
        print(f"-> {name}: {ipv6}")
```

```
        found = True
```

```
if not found:
```

```
    print("No matching restaurants found.")
```

Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS:

LAB REPORT: AI-GENERATED IPv6 SUBNETTING & MANUAL INTERFACE ASSIGNMENT

1. AI-GENERATED IPv6 SUBNETTING SCENARIO (Source: ChatGPT)

Scenario: A university campus network has been allocated a base IPv6 prefix of **2001:db8:cafe::/48**. The network administrator needs to divide this prefix into smaller subnets to accommodate at least 3 distinct campus departments:

1. Faculty of Engineering
2. Student Dormitories
3. Admin & Library Services

To achieve this, the administrator decides to subnet down from the **/48** base prefix to a **/64** prefix boundary (standard subnet size for LAN segments), which yields a total of $2^{16} = 65,536$ available subnets.

Subnet Allocation Plan:

- Base Prefix: 2001:db8:cafe::/48
- Subnet Mask (Prefix): /64 (Leaves 64 bits for interface IDs)

Department Subnets:

1. Faculty of Engineering: 2001:db8:cafe:1::/64
2. Student Dormitories: 2001:db8:cafe:2::/64
3. Admin & Library Services: 2001:db8:cafe:3::/64
4. MANUAL CONFIGURATION STEPS

Objective: Manually assign a valid static IPv6 address from the "Faculty of Engineering" subnet (2001:db8:cafe:1::/64) to the primary system network interface.

--- STEP A: LINUX SYSTEM CONFIGURATION ---

1. Open a terminal with superuser privileges.
2. Identify the active network interface name (e.g., eth0 or enp3s0): `$ ip link show`
3. Assign a static IPv6 host address from Subnet 1 (e.g., 2001:db8:cafe:1::100/64): `$ sudo ip -6 addr add 2001:db8:cafe:1::100/64 dev eth0`
4. Verify the binding using the ip command: `$ ip -6 addr show dev eth0`

--- STEP B: WINDOWS SYSTEM CONFIGURATION ---

1. Open Command Prompt or PowerShell as Administrator.
2. Assign the static IPv6 address to the primary adapter (e.g., "Ethernet"):
`netsh interface ipv6 add address "Ethernet" 2001:db8:cafe:1::100/64`
3. Verify the configuration:
`ipconfig`
4. VERIFICATION & PING TEST

To test connectivity within the newly assigned subnet, ping the gateway or another host within 2001:db8:cafe:1::/64:

- Linux: `$ ping -6 2001:db8:cafe:1::1`
- Windows: `> ping -6 2001:db8:cafe:1::1`